Photometric properties and evidence of duplicity for SZ Lyncis

Thomas G. Barnes III and Thomas J. Moffett

Department of Astronomy and McDonald Observatory, The University of Texas at Austin, Austin, Texas 78712

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UBV photometry with a time resolution of 20 sec is reported for the AI Vel variable, SZ Lyn. The reported irregularities in height of maximum light are confirmed and found to have a total amplitude ~0.04 mag. No periodic amplitude modulation is observed. It is demonstrated that the colors of SZ Lyn are normal for its spectral type and luminosity class. The periodic variation from a linear ephemeris found by van Genderen is confirmed and its parameters improved. The most plausible interpretation of this periodicity is a light-travel-time effect, leading to the prediction that SZ Lyn is a single-lined spectroscopic binary with period 3.14 yr and total velocity amplitude 19 km/sec.

INTRODUCTION

The ultrashort-period variable SZ Lyncis was discovered by Hoffmeister (1949) and classified as a W Ursa Majoris system with a period of 0.275 day by Tsesevich (1956a,b). Subsequent photometry by Schneller (1961) and Eggen (1962a) showed the true period to be near 0.12 day, leading them to suggest a reclassification as a short-period RR Lyrae variable. However, Broglia (1963) pointed out that the large light amplitude ($\Delta V \sim 0.5$ mag) and asymmetric light curve were more consistent with the dwarf Cepheid (AI Vel) class, as SZ Lyn is now designated.

The nature of the periodicity has been examined by Broglia (1963) and van Genderen (1963, 1967) with interesting results. In his second paper, van Genderen argued that the time of maximum residuals from a linear ephemeris followed a sine wave of period 1129 days. But because only 1.6 cycles of the sine wave were encompassed by the observations, the reality of it remained somewhat unclear, particularly since Joshi and Srivastava (1967) and Wisse and Wisse (1969) reported three times of maximum which deviated substantially from the sinusoidal ephemeris.

Irregularities in the light curve have been claimed by some observers and denied by others. Broglia (1963), van Genderen (1963), and Binnendijk (1968) found no evidence for variation in the amplitude or shape of the light curve. On the other hand, van Genderen (1967), Joshi and Srivastava (1967), and Wisse and Wisse (1969) reported amplitude irregularities of about ± 0.03 mag accompanied by changes in the form of the light curve.

We have observed SZ Lyn with the McDonald Observatory's high-speed photometer in an attempt to resolve the above discrepancies.

I. OBSERVATIONS

All observations were obtained with the 91-cm reflector at McDonald Observatory using the high-speed, pulse-counting photometer described by Nather and Warner (1971) and Owen *et al.* (1972). The photomultiplier used was a specially selected Amperex 56DVP (a blue-sensitive, bi-alkali photocathode). A standard set of *UBV* filters was employed. The integration time was 5 sec per filter, yielding a cycle time of 20 sec (a fourth filter position contained no filter and was not used in the reductions).

The standard techniques of differential photometry were employed both in the observation and reduction procedures, except that the comparison star was observed only at 30-min intervals. We used the same comparison star as did van Genderen (1963), BD+45°1544, and adopted his magnitude and colors: $V=9.43$, $(B-V)=+0.46$, $(U-B)=+0.03$.

Transformation to the *UBV* system was accomplished by mean transformation coefficients and zero points adjusted to give the above values for BD+45° 1544. Therefore our transformation to the *UBV* system is estimated to be uncertain in zero point by ± 0.03 mag in V and ± 0.02 mag in the colors.

On the other hand, the internal precision of our results is very high. Judged from the repeatability of the measures, the standard deviation is ± 0.007 mag in V, ± 0.007 mag in $(B-V)$, and ± 0.005 mag in $(U-B)$ for the 5-sec integrations.

Four individual cycles of SZ lyn were observed, at least in part. During the first cycle no comparison star observations were obtained, so that cycle is used here only for the time of maximum.

The times of maximum brightness in the B bandpass were determined visually to an estimated accuracy of ± 0.00046 day (40 sec). They are the following: $JD_{\odot}$ 2 441 678.84753, 2 441 683.79092, 2 441 683.9096, and 2 441 684.03119.

The individual observations were collected in phase using the period determined below. Normal points were computed for the light and color curves by averaging the several cycles together into phase bins $0.02P$ wide (~208 sec). Thus, for each cycle contributing to a given normal point, approximately ten 5-sec integrations were summed. We estimate that the normal points are differentially precise (standard error) to ± 0.002 mag in V and in the colors, although these precisions vary somewhat with phase as a result of the varying number of cycles averaged at each phase. The light and color variations are shown in Figs. 1 and 2. Normal points are given in Table I.

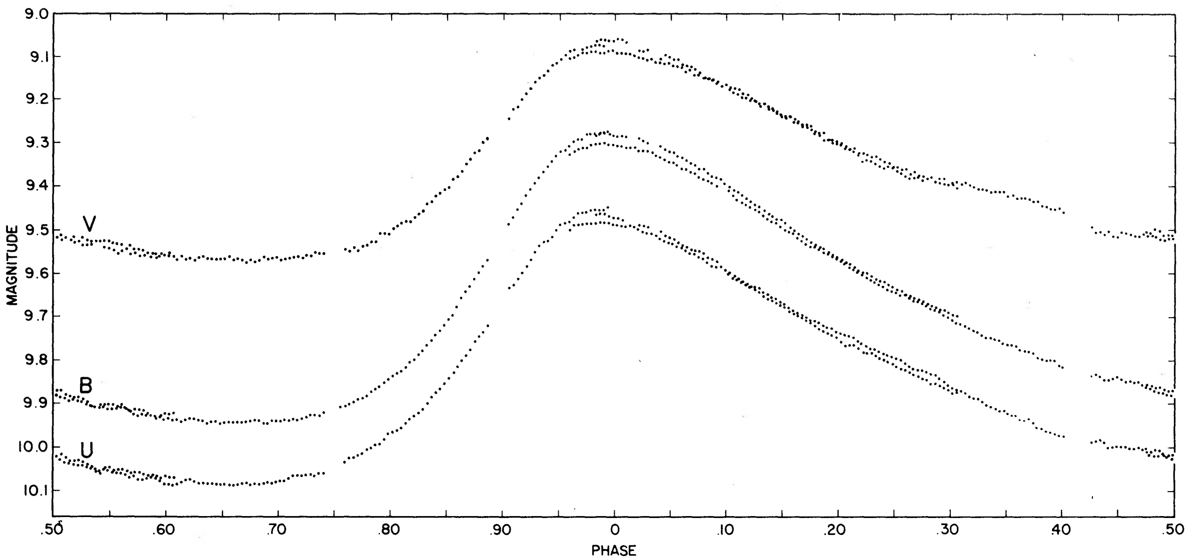

FIG. 1. The UBV magnitudes versus phase.

(Note that all figures and discussion of the light variations use *visual* maximum as phase zero, whereas for compatibility with previous observers, the period analyses use *blue* maximum as phase zero.)

The values at maximum and minimum light from the normal curves are the following:

Maximum, $V=9.079$, $(B-V)=0.220$, $(U-B)=0.191$.
Minimum, $V=9.536$, $(B-V)=0.368$, $(U-B)=0.151$.

II. PERIOD ANALYSIS

Listed in Table II are all published times of maximum for SZ Lyn. We assigned a relative weight of 0.1 to all visual observations, 1.0 for all photoelectric observations, and 3.0 for our high-time-resolution results. A least-squares solution using all 94 times of maximum resulted in the following linear ephemeris:

$$T_{\max}(\mathrm{JD}_\odot)=2\,438\,124.39770+0.12053481E, \quad (1)$$
$$\pm 56 \text{ (rms)} \qquad 4 \text{ (rms)}$$

where E is the number of elapsed periods, adopting van Genderen's (1967) zero point. The residuals $(O-C)_1$, determined from Eq. 1 are given in the fourth column of Table II. As first pointed out by van Genderen (1967), we also found that a linear ephemeris produced $(O-C)$'s which appear to follow a sine-curve relation.

We therefore fit the epochs of maximum to the following relation:

$$T_{\max}(\mathrm{JD}_\odot)=T_0+EP_0-A\cos 2\pi\left(\frac{EP_0}{P_1}-\varphi\right), \quad (2)$$

where T_0 is the initial epoch of maximum corresponding to $E=0$, P_0 is the pulsation period of SZ Lyn, A is the semiamplitude of the sine curve, P_1 is the period of the sinusoid, and φ is a phase shift.

Values for the five parameters T_0, P_0, A, P_1, and φ were determined by the method of least squares (differential corrector). Since the timing accuracy of

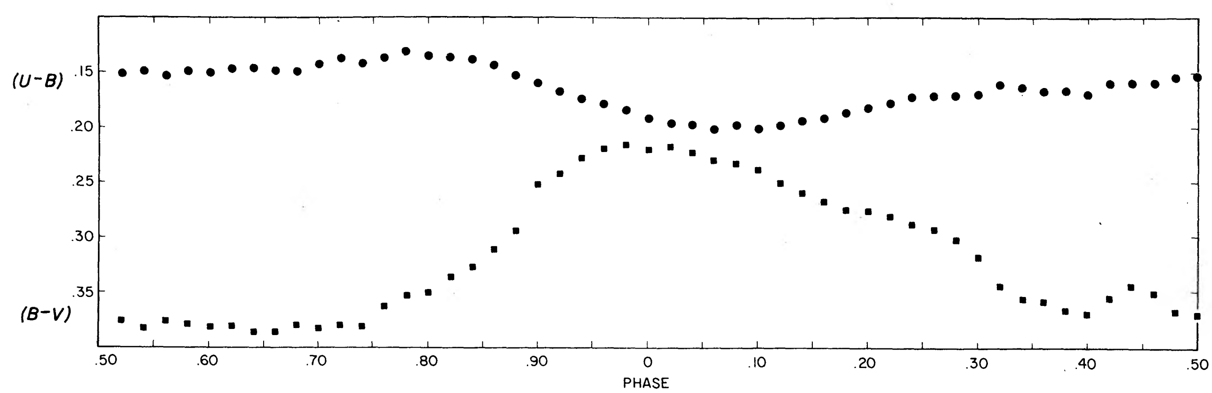

FIG. 2. The color curves (normal points) versus phase.

TABLE I. Normal points of SZ Lyncis.

Phase	V	$(B-V)$	$(U-B)$
0.0	9.079	0.220	0.191
0.02	9.096	0.217	0.198
0.04	9.113	0.222	0.197
0.06	9.126	0.229	0.201
0.08	9.151	0.232	0.197
0.10	9.177	0.238	0.200
0.12	9.202	0.250	0.197
0.14	9.229	0.259	0.193
0.16	9.255	0.267	0.190
0.18	9.281	0.274	0.185
0.20	9.310	0.275	0.181
0.22	9.337	0.280	0.177
0.24	9.360	0.287	0.171
0.26	9.380	0.292	0.170
0.28	9.394	0.301	0.170
0.30	9.405	0.317	0.169
0.32	9.413	0.343	0.160
0.34	9.424	0.354	0.162
0.36	9.439	0.357	0.165
0.38	9.454	0.365	0.165
0.40	9.469	0.368	0.168
0.42	9.509	0.354	0.158
0.44	9.525	0.343	0.157
0.46	9.527	0.349	0.157
0.48	9.526	0.366	0.152
0.50	9.536	0.368	0.151
0.52	9.545	0.375	0.151
0.54	9.553	0.383	0.149
0.56	9.565	0.376	0.153
0.58	9.575	0.379	0.149
0.60	9.581	0.381	0.150
0.62	9.589	0.381	0.147
0.64	9.589	0.386	0.146
0.66	9.589	0.386	0.148
0.68	9.593	0.380	0.149
0.70	9.589	0.383	0.143
0.72	9.584	0.380	0.137
0.74	9.575	0.381	0.142
0.76	9.570	0.363	0.137
0.78	9.556	0.353	0.131
0.80	9.519	0.350	0.135
0.82	9.488	0.336	0.136
0.84	9.440	0.327	0.138
0.86	9.389	0.311	0.143
0.88	9.325	0.294	0.153
0.90	9.242	0.252	0.159
0.92	9.194	0.242	0.167
0.94	9.137	0.228	0.173
0.96	9.101	0.219	0.178
0.98	9.087	0.215	0.183

visual observations is poor, only photoelectric observations were included in the solution. The reported times of maximum at $E=2919$, 2977, and 18 459 which showed very large residuals were excluded from the differential-corrector fit.

Table III compares our results on the period of SZ Lyn with those of van Genderen. Our values for the five parameters agree very well with his except for the period of the sinusoid, for which we find 1146^{d} rather than 1129^{d}. The observations used by van Genderen only covered $1.6P_1$, whereas we extended the baseline of coverage to $3.8P_1$ resulting in an improved period. We believe that any doubt about the reality of the sinusoidal variation is now removed.

Figure 3 indicates the goodness of fit for the nonlinear ephemeris. The ordinate $(O-C)$ is the residual from the linear portion of Eq. 2, and the abscissa Ψ is the phase of the sinusoid. The $(O-C)_2$'s determined from Eq. 2 and the phase of the sinusoid Ψ are given in columns 5 and 6 to Table II.

The three maxima excluded from the fit to Eq. 2 are shown as Xs in Fig. 3. These were previously reported by Joshi and Srivastava (1967) and by Wisse and Wisse (1969) to deviate from van Genderen's nonlinear ephemeris. The points deviate from our nonlinear ephemeris by four times the stated timing uncertainties given by the above authors. From the data at hand we cannot choose between the alternate hypotheses that the deviations result (1) from real effects at the star or (2) from observational error in the timing.

III. DISCUSSION

A. Light Curves

The magnitude curves in Fig. 1 show the usual characteristics associated with radial pulsation in the fundamental mode for stars in the Cepheid instability strip. There is a rapid rise to maximum, a slow decline to minimum, and nearly flat minimum centered on phase 0.68. The V light curve exhibits a "hump" prior to a minimum, similar to that seen for some Cepheid-type variables. Finally, a progressive delay in time of maximum occurs with increasing wavelength. The delay amounts to $\sim 0.02P$ between U and V. This is generally attributed to the phase difference between the radius and temperature variations, although to our knowledge it has not previously been reported for such a short-period variable.

As other observers have reported, we also find cycle-to-cycle variations in the magnitude achieved at

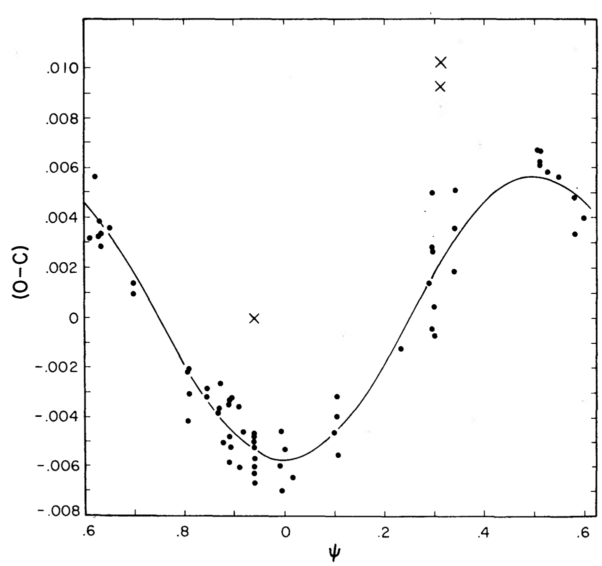

FIG. 3. The residuals from the linear ephemeris in Eq. 2 versus phase in the 1146-day cycle. The smooth curve is the fitted sinusoid.

TABLE II. Observed maxima.

Epoch of maximum JD⊙+ 2400000	E	W	$(O-C)_1$	$(O-C)_2$	ψ	Max B	Observer*
33357.26	−39550	0.1	0.01	0.02	0.839		So
33392.20	−39260	0.1	−0.00	0.01	0.870		So
33946.549	−34661	0.1	0.008	0.007	0.354		Ts
33947.607	−34652	0.1	−0.018	−0.019	0.355		Ts
34034.18	−33934	0.1	0.01	0.01	0.430		So
34041.522	−33873	0.1	−0.000	−0.003	0.437		Ts
34064.437	−33683	0.1	0.013	0.010	0.457		Ts
34414.20	−30781	0.1	−0.02	−0.01	0.762		So
34440.359	−30564	0.1	−0.013	−0.009	0.785		Ts
34445.21	−30524	0.1	0.02	0.02	0.789		So
35592.339	−21007	0.1	0.016	0.019	0.790		Ts
35593.392	−20998	0.1	−0.016	−0.013	0.791		Ts
37367.441	− 6280	1.0	0.002	−0.001	0.339		Sch
37368.288	− 6273	1.0	0.005	0.002	0.340		Sch
37368.407	− 6272	1.0	0.004	0.000	0.340		Sch
37642.021	− 4002	1.0	0.004	−0.001	0.579		E
37780.031	− 2857	1.0	0.001	−0.000	0.699		H and X
37780.152	− 2856	1.0	0.002	−0.000	0.700		H and X
37977.343	− 1220	1.0	−0.002	0.002	0.872		H and X
37997.351	− 1054	1.0	−0.003	0.001	0.889		H and X
38001.568	− 1019	1.0	−0.005	−0.001	0.893		vG
38001.689	− 1018	1.0	−0.004	−0.000	0.893		vG
38003.378	− 1004	1.0	−0.003	0.001	0.894		H and X
38021.576	− 853	1.0	−0.006	−0.001	0.910		vG
38021.699	− 852	1.0	−0.003	0.001	0.910		vG
38032.305	− 764	1.0	−0.004	0.001	0.920		H and X
38053.276	− 590	1.0	−0.006	−0.001	0.938		H and X
38055.205	− 574	1.0	−0.006	−0.001	0.940		H and X
38055.3268	− 573	1.0	−0.0045	0.0005	0.940	9.338	B
38055.4463	− 572	1.0	−0.0055	−0.0006	0.940	9.331	B
38055.5681	− 571	1.0	−0.0042	0.0007	0.940	9.332	B
38057.3752	− 556	1.0	−0.0051	−0.0002	0.942	9.335	B
38057.4964	− 555	1.0	−0.0045	0.0005	0.942	9.349	B
38057.6172	− 554	1.0	−0.0042	0.0007	0.942	9.329	B
38090.2803	− 283	0.1	−0.0060	−0.0009	0.970		L
38114.388	− 83	1.0	−0.005	−0.000	0.991	9.354	B
38118.367	− 50	1.0	−0.004	0.001	0.995	9.340	vG
38118.485	− 49	1.0	−0.006	−0.001	0.995	9.345	vG
38124.393	0	1.0	−0.005	0.000	0.000		vG
38140.423	133	1.0	−0.006	−0.001	0.014	9.346	B
38144.281	165	0.1	−0.005	0.000	0.017		L
38144.395	166	0.1	−0.011	−0.006	0.017		L
38145.354	174	0.1	−0.017	−0.012	0.018		L
38150.309	215	0.1	−0.004	0.001	0.023		L
38152.354	232	0.1	−0.008	−0.003	0.024		L
38155.362	257	0.1	−0.013	−0.008	0.027		L
38159.3395	290	0.1	−0.0133	−0.0084	0.030		L
38160.30	298	0.1	−0.02	−0.01	0.031		L
38161.3949	307	0.1	−0.0070	−0.0021	0.032		L
38162.3538	315	0.1	−0.0124	−0.0075	0.033		L
38163.3132	323	0.1	−0.0172	−0.0124	0.034		L
38164.2782	331	0.1	−0.0165	−0.0116	0.035		L
38460.21	2786	1.0	0.00	−0.00	0.293		J and S
38463.342	2812	1.0	0.000	−0.002	0.296		J and S
38464.189	2819	1.0	0.004	0.001	0.297		J and S
38465.276	2828	1.0	0.006	0.003	0.298		J and S
38466.238	2836	1.0	0.004	0.001	0.298		J and S
38467.200	2844	1.0	0.001	−0.002	0.299		J and S
38471.297	2878	1.0	0.000	−0.003	0.303		J and S
38476.249	2919	1.0	0.010	0.007	0.307		J and S
38483.241	2977	1.0	0.011	0.008	0.313		J and S
38701.8878	4791	1.0	0.0078	0.0011	0.504		Bin.
38708.6377	4847	1.0	0.0078	0.0011	0.510	9.354	vG
38709.6014	4855	1.0	0.0072	0.0005	0.511	9.351	vG
38712.6148	4880	1.0	0.0072	0.0005	0.513	9.341	vG
38725.5117	4987	1.0	0.0069	0.0003	0.525		vG
38752.5114	5211	1.0	0.0068	0.0004	0.548	9.354	vG
38788.5505	5510	1.0	0.0060	0.0001	0.580	9.359	vG
38809.4022	5683	1.0	0.0052	−0.0004	0.598	9.339	vG
38830.4949	5858	1.0	0.0043	−0.0009	0.616	9.340	vG
38834.234	5889	1.0	0.007	0.0017	0.620		J and S

TABLE II (*continued*)

Epoch of maximum JD⊙+ 2400000	E	W	$(O-C)_1$	$(O-C)_2$	ψ	Max B	Observer*
38844.4771	5974	1.0	0.0044	−0.0004	0.628	9.354	vG
38849.5396	6016	1.0	0.0045	0.0002	0.633	9.344	vG
38850.3839	6023	1.0	0.0050	0.0003	0.634	9.338	vG
38850.5034	6024	1.0	0.0040	−0.0007	0.634	9.355	vG
38871.3567	6197	1.0	0.0048	0.0006	0.652	9.345	vG
39052.5129	7700	1.0	−0.0028	−0.0017	0.810	9.336	vG
39052.6355	7701	1.0	−0.0008	0.0003	0.810	9.341	vG
39054.5631	7717	1.0	−0.0017	−0.0006	0.812	9.335	vG
39054.6846	7718	1.0	−0.0008	0.0004	0.812	9.340	vG
39092.7729	8034	1.0	−0.0015	0.0007	0.845		Bin
39092.8931	8035	1.0	−0.0018	0.0004	0.845		Bin.
39121.7004	8274	1.0	−0.0023	0.0005	0.870		Bin.
39121.8208	8275	1.0	−0.0025	0.0004	0.871		Bin.
39130.4981	8347	1.0	−0.0037	−0.0007	0.878	9.316	vG
39145.4461	8471	1.0	−0.0020	0.0013	0.891	9.334	vG
39145.5642	8472	1.0	−0.0044	−0.0012	0.891	9.345	vG
39205.4706	8969	1.0	−0.0038	0.0002	0.944	9.342	vG
39531.401	11673	1.0	0.000	−0.0008	0.228		J and S
40349.3521	18459	1.0	0.0023	0.0054	0.942	9.386	W and W
41678.8475	29489	3.0	−0.0012	−0.0003	0.102		B and M
41683.7909	29530	3.0	0.0003	0.0011	0.106	9.292	B and M
41683.9091	29531	3.0	−0.0021	−0.0013	0.107	9.312	B and M
41684.0312	29532	3.0	−0.0005	0.0003	0.107	9.278	B and M

* So = Soloviev (1959b), Ts = Tsesevich (1956a), Sch = Schneller (1961), E = Eggen (1962a), H and X = He Tian-jian and Xiong Da-run (1964), vG = van Genderen (1963, 1967), B = Broglia (1963), J and S = Joshi and Srivastava (1967), Bin. = Binnendijk (1968), W and W = Wisse and Wisse (1969), L = Lange (quoted in Tsesevich 1969, p. 319), B and M = Barnes and Moffett.

maximum. The effect can be seen in Fig. 1. The total variation in all three bandpasses is 0.04 mag. This confirms the suppositions of van Genderen (1967).

However, we do not confirm the radical change in shape of maximum claimed by van Genderen (1967) and Wisse and Wisse (1969). These authors remark that the brighter maxima are much more sharply peaked than the fainter ones. The three cycles observed by us appear to have nearly the same shape.

Many of the variables classified as AI Vel type exhibit periodic variations in the height of maximum with a period $\sim 3P_0$–$4P_0$. Gefferth and Szeidl (1962) searched for this effect in SZ Lyn without success. Using the more extensive photometry now available (Table II), we repeated the search. No periodicities in height of maximum were found. In particular, Fig. 4 shows that no significant variation in maximum occurs with the period of 1146 days. We shall remark further on this later.

B. Color Curves

The $(U-B)-(B-V)$ loop traversed by SZ Lyn during its cycle is shown in Fig. 5 with the Hyades main sequence according to Eggen (1962b). The overall appearance is similar to the loops of those other AI Vel variables which are not complicated by pronounced beat phenomena, e.g., YZ Boo (Heiser and Hardie 1964), AD CMi (Abhyankar 1959), and DY Her (Hardie and Lott 1969). In all these cases the loops are quite narrow in $(U-B)$ and oriented nearly parallel to the main sequence. Bessell (1969) has emphasized that the narrow loop indicates a high surface gravity, relative to variables with wider loops, such as RR Lyr.

That SZ Lyn lies below the Hyades relation is presumably due in part to interstellar reddening. Eggen (1962a) compared the UBV colors of SZ Lyn to those of AD CMi, which lies somewhat above the Hyades relation. On the assumption that AD CMi is un-

TABLE III. Times of maximum parameters.

Source	No. Max.	T_0	s.e.	P_0	s.e.	P_1	s.e.	A	s.e.	φ	s.e.
				(a) Linear ephemeris							
van Genderen	60	2 438 124.39714	0.00066	$0^d.12053487$	$0^d.00000010$						
Present study	94	2 438 124.39770	0.00056	0.12053481	0.00000004						
				(b) Linear plus sinusoidal ephemeris							
van Genderen	48	2 438 124.39849	0.00021	0.12053493	0.00000005	1129^d	18^d	$0^d.00569$	0.00018	−0.019	0.011
Present study	66	2 438 124.39828	0.00017	0.120534906	0.000000020	1146	10	0.00572	0.00019	−0.010	0.008

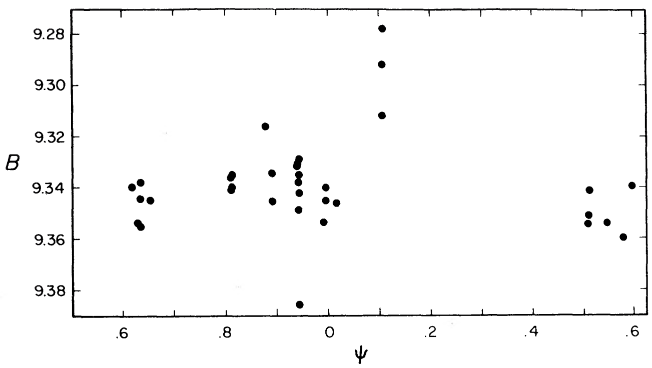

FIG. 4. The B magnitude at maximum light versus phase in the 1146-day cycle.

reddened, he deduced $E(B-V)=0.13$ mag for SZ Lyn. The comparison star BD +45° 1544 is only 18 arcmin distant from SZ Lyn and has very nearly the same apparent magnitude. With the assumption that it is a main-sequence star, van Genderen (1963) estimated $E(B-V)<0.05$ mag, which he then applied to SZ Lyn. As indicated in Fig. 5 we find $E(B-V)=0.04$ mag for BD +45° 1544. We also point out that this should be a lower limit to the reddening of SZ Lyn on the grounds that it is more distant than an F5 main-sequence star of the same apparent magnitude. Another estimate of the color excess can be obtained from Parenago's relation, as calibrated by Sharov (1964). Adopting an absolute magnitude of +2 mag, we obtain $E(B-V)=0.08$ mag. (If SZ Lyn were entirely above the obscuring material, we would obtain $E(B-V)=0.10$ mag.) We conclude that the color excess is approximately 0.06 mag.

Unreddened by this amount, the centroid of the loop falls just below the Hyades relation. Thus we do not agree with van Genderen's (1963) suggestion that the depression is nearly all attributable to some cause other than interstellar reddening, such as unusually strong line blanketing in the U bandpass. This latter possibility is further rendered unlikely by the Preston ΔS values observed by Alania (1972). From eight spectra near maximum, he found $\Delta S=1\pm1$, indicating a possible metal *deficiency* for SZ Lyn compared to the Hyades.

The remaining small depression below the main-sequence curve can be attributed to a lower surface gravity for SZ Lyn. This is consistent with spectral classification of several AI Vel stars as luminosity class III or IV (Bessell 1969).

An unusual feature in the color loop of SZ Lyn occurs during the phase interval 0.32–0.44 and warrants

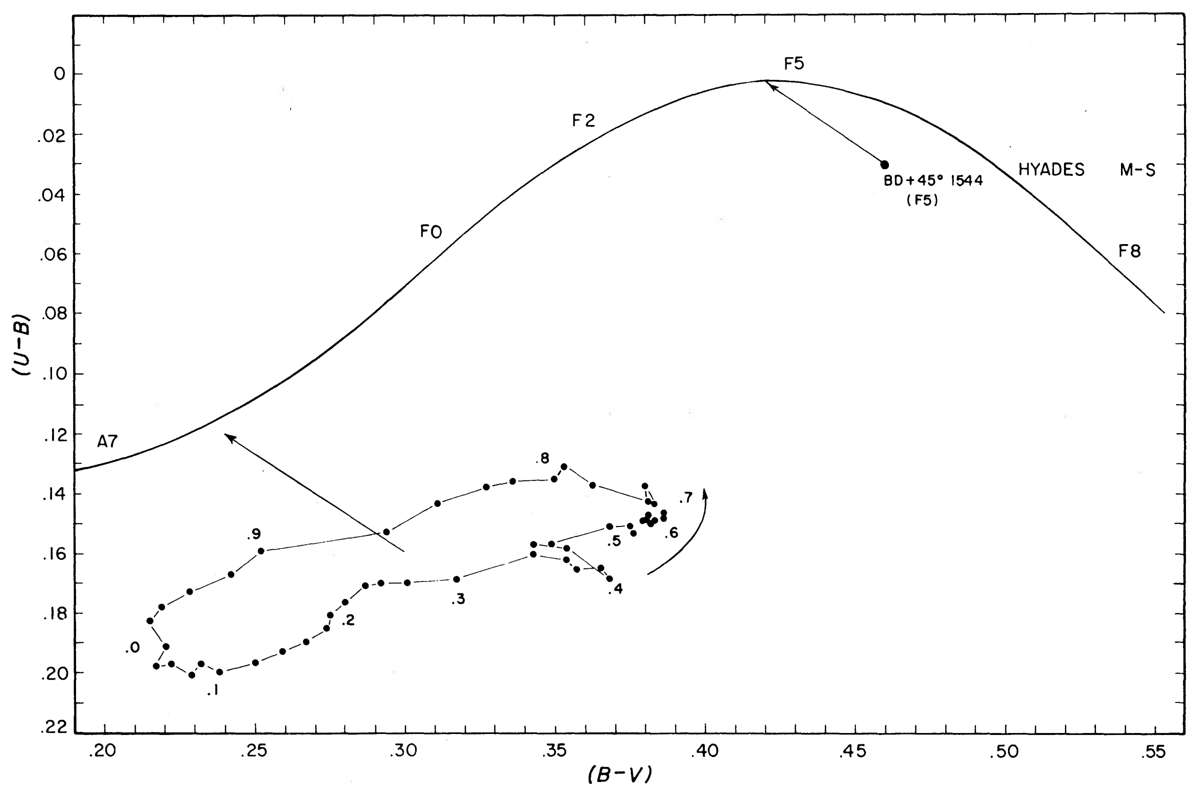

FIG. 5. The two-color diagram for SZ Lyn during the mean cycle. The direction of motion is indicated by the arrow and phases are marked for each $0.1P$. Also shown are the displacements resulting from removal of a color excess of 0.04 mag from the comparison star and 0.06 mag from the variable.

comment. The feature appears in Fig. 5 as a spur on the otherwise smooth loop. It is clearly associated with the hump in the visual light curve and arises because of the absence of such a hump in the B and U light curves. Unfortunately, for most of this phase interval we have coverage for only one cycle, so we cannot state that this behavior reproduces each cycle, nor even that it is definitely a real stellar phenomenon.

C. Radius Determination

Using the radial velocities of SZ Lyn given by Woolley and Aly (1966), we attempted an application of Wesselink's (1946) method to determine the absolute radius. The method did not produce a unique solution, even when modified as suggested by Abt (1959) for RR Lyr variables. This was not unexpected since to our knowledge the method has never been successfully applied to any AI Vel variable (for example, see Abhyankar 1959).

D. Time of Maximum Variation

Two interpretations are available for the 3.14-yr periodicity in the time of maximum light: (1) simultaneous excitation of two or more pulsation frequencies causing a beat phenomenon, and (2) modulation of the phase of maximum light by binary motion. We reject the first alternative because no beat phenomena are observed in the light curve.

In the second hypothesis, the variation in time of maximum light is attributed to the light travel time across the orbit of SZ Lyn in a binary. For a circular orbit of radius a_{SZ} and inclination i, the time of light maximum is given by

$$T_{\max}=T_0+EP_0-\Delta t\cos 2\pi\left(\frac{EP_0}{P_B}-\varphi\right), \qquad (4)$$

where P_0 is the pulsation period, P_B is the orbital period, and $\Delta t=(a_{SZ}\sin i)/c$. This is just Eq. 2 with $\Delta t=A$ and $P_B=P_1$. We are therefore justified in taking P_1 and A from Table III to be the oribtal period and projected light travel time, respectively.

Consequently, within the observational uncertainties, the long-term variation in time of maximum is mathematically equivalent to a single-lined spectroscopic binary with parameters:

$$e=0 \text{ (adopted)};$$

$$P=3.14\pm0.03 \text{ yr};$$

$$a_{SZ}\sin i=c\Delta t=0.99\pm0.03 \text{ AU};$$

$$K_{SZ}=\frac{2\pi a_{SZ}\sin i}{P}=9.4\pm0.4 \text{ km/sec};$$

$$f(\mathfrak{M}_2)=\frac{\mathfrak{M}_2^3\sin^3 i}{(\mathfrak{M}_1+\mathfrak{M}_2)^2}=\frac{(c\Delta t)^3}{P^2}=0.098\pm0.010\ \mathrm{M}_\odot,$$

where SZ Lyn has mass $\mathfrak{M}_1$. As these values are not unreasonable for a binary and bearing in mind the difficulties of the alternate hypotheses, we adopt for further discussion the interpretation that the variation in time of maximum is caused by a light travel effect in a binary.

E. Binary Model

The possibility that SZ Lyn is a spectroscopic binary immediately leads to the hope that a mass can be determined. This is particularly important because the masses of the AI Vel variables (as well as the physical reality of the group) are an intensely controversial topic. Bessell (1969) and Petersen and Jorgensen (1972) have argued for masses near $0.5\mathrm{M}_\odot$ and an advanced evolutionary state. Eggen (1970) and Baglin *et al.* (1973) have concluded that the AI Vel stars are in fact merely large amplitude δ Sct variables, therefore with masses near $1.5\mathrm{M}_\odot$ and in a much-less-advanced evolutionary state.

In Fig. 6 we show the relationship between the masses of the two components for assumed values of the orbital inclination. The curve for $i=90°$ sets a lower limit to $\mathfrak{M}_2$ at each $\mathfrak{M}_1$.

A further restriction is placed upon the permitted region by the spectroscopic observations of Alania (1972) and Woolley and Aly (1966). Alania examined eight spectra of SZ Lyn taken at 166 Å/mm for ΔS measures and yet did not remark upon spectral duplicity. Woolley and Aly measured 17 spectra taken at 67 Å/mm for radial velocities and also failed to comment upon doubled lines. We must conclude that the companion is at least 1–2 mag fainter than SZ Lyn.

Most investigators have assigned absolute magnitudes from 1–4 mag to the AI Vel variables with the consensus that $M_v\sim2$ mag is a reasonable estimate [see Baglin *et al.* (1973) for a summary]. Hence, the companion must be fainter than about $M_v\sim3.5$ mag. We conclude that the companion is either a white dwarf or much more probably a main-sequence star later than about F5 and therefore of mass less than $\sim1.4\mathrm{M}_\odot$. This places an upper limit to the permitted mass region in Fig. 6.

For this system it is clear that we cannot limit further the mass of SZ Lyn without detection of the companion. During the next observing season a careful spectrographic search for evidence of duplicity should be undertaken.

As a final remark on the binary model, we stress that, while it is to us the most plausible interpretation of the variable time of maximum, it remains only a reasonable hypothesis until confirming data are available. The simplest and most easily accomplished test is to examine the radial velocity of SZ Lyn for the 18.8-km/sec amplitude predicted by the model. One set of radial velocity measures appears in the literature, those by Woolley and Aly, but they span only ten days.

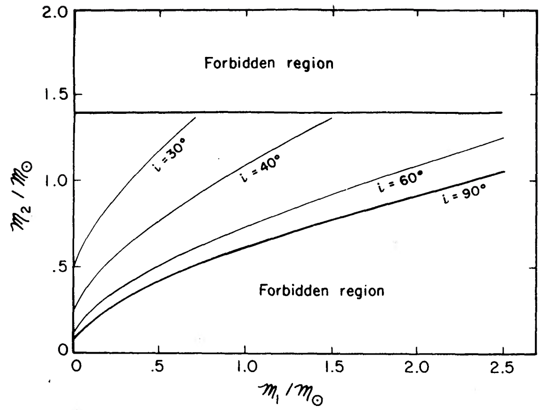

FIG. 6. Solutions to the mass function given in Eq. 6 for various choices of $\sin^3 i$.

(Fortuitously, their observations occurred near a binary phase at which SZ Lyn is predicted to have the gamma velocity of the system.) Care will have to be taken to account for the 24-km/sec amplitude associated with 0.12-day period.

IV. SUMMARY

By means of high-time-resolution photometry, we have further defined the nature of variability for the AI Vel star, SZ Lyn. The light curve shows the usual Cepheid characteristics, including a "hump" on the declining branch of the V curve and a progressive delay in time of maximum with increasing wavelength. Cycle-to-cycle irregularities are present in the light curves, being larger near maximum than near minimum. At maximum, the three cycles we observed differed by a total of 0.04 mag in V. The radical changes in shape of light curve reported by some observers are not confirmed. A search for periodic amplitude modulation was negative.

The $(U-B)-(B-V)$ loop of SZ Lyn is similar to those of other AI Vel variables in shape and location. A discussion of the reddening of SZ Lyn indicates a most probable value near 0.06 mag for $E(B-V)$. From this we conclude that the location of SZ Lyn in the color–color diagram is consistent with its spectral type and luminosity class. It is not necessary to invoke unusually strong metallic line blanketing to account for its location below the main-sequence curve, as others have suggested.

The periodic variation in time of maximum light discovered by van Genderen (1967) is here confirmed. The variation has period 1146±10 days and semi-amplitude 494±16 sec. Several hypotheses are considered to explain this effect, with the most plausible being that SZ Lyn has an unseen companion. This interpretation is augmented by the absence of amplitude modulation in the light curve with the 1146-day period. From the available data the companion is inferred to be a main-sequence star later than F5.

An attempt to apply Wesselink's method to determine the absolute radius of SZ Lyn failed.

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